

## RESEARCH ARTICLE

# Exploring Factors Involved in Loan Approval Decision: Deep Insights and Data Analytics Techniques

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**ABSTRACT** Accurate yet transparent credit-risk models are essential for responsible lending in the face of tightening global AI regulations. We propose an end-to-end, reproducible pipeline for loan-default prediction that integrates a three-way consensus feature-selection ensemble using VarianceThreshold, RFE with logistic regression and XGBoost gain ranking; a lightweight one-dimensional convolutional neural network optimised for tabular data; post-hoc explainability via KernelSHAP embedded directly in the inference loop; and continuous system-level profiling of CPU, RAM, GPU and latency. Using the public GiveMeSomeCredit dataset, our method reduces the original 11-feature space to a stable subset of five predictors, achieving a ROC-AUC of 0.862 and an F1-score of 0.55 on a stratified 20% hold-out, surpassing the logistic regression and XGBoost baselines by 9% and 4% ROC-AUC, respectively. Ablation analysis reveals that the consensus feature selection contributes 57 percent of the total accuracy gain, while the 1D-CNN architecture contributes an additional 38 percent. Fairness assessment shows disparate-impact and equal-opportunity gaps below 5 percent across gender and age cohorts, aligning with emerging EU AI Act thresholds. End-to-end inference averages 18 milliseconds on CPU-only hardware, confirming real-time viability. All code, trained models, evaluation artifacts and resource logs are openly archived, offering a deploy-ready blueprint for lenders aiming to modernise legacy scorecards without sacrificing interpretability, compliance, or operational efficiency.

**INDEX TERMS** Loan-default prediction, credit-risk modeling, 1D-CNN, feature selection, explainability.

## I. INTRODUCTION

Credit-Risk assessment lies at the heart of modern consumer finance. Every loan application forces lenders to weigh the prospect of profit against the possibility of default and even marginal improvements in prediction translate into sizeable monetary impacts at portfolio scale. Traditional credit-scoring systems which are built on logistic regression or simple decision trees and hand-crafted feature engineering that remain popular because they are inexpensive, fast and reasonably transparent but Yet their capacity to capture non-linear interactions among the high-dimensional attributes now routinely collected by banks, fintechs and credit bureaus is inherently limited [1], [2], [3]. As a result,

they often underperform in volatile economic climates or when borrower profiles deviate from the historical norm.

Recent advances in deep learning offer a path to more accurate credit risk prediction, but two barriers hinder adoption in regulated domains: class imbalance, since defaults are rare and explainability, as closed box models are difficult to justify to auditors and regulators [4], [5]. Regulatory bodies such as the European Banking Authority and the U.S. Consumer Financial Protection Bureau are tightening transparency requirements, while practitioners must also meet operational constraints such as real-time speed, hardware efficiency and reproducibility [6], [7], [8]. To address these challenges, we propose a comprehensive pipeline for default risk prediction using the *Give Me Some Credit* dataset [9]. The framework combines ensemble feature selection (VarianceThreshold, RFE, XGBoost),

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a lightweight one-dimensional CNN for efficient deployment and SHAP-based explanations that provide both global rankings and per-applicant insights [10], [11], [12], [13].

Beyond predictive performance and interpretability, we address practical engineering considerations that are often overlooked in academic work but significantly influence industrial decision-making. The pipeline incorporates the system-level profiling, tracking memory footprint, CPU/GPU utilization and per-sample inference latency to facilitate capacity planning for real-time deployment [14]. All artifacts, including trained models, selected features, evaluation plots, SHAP visualizations and resource logs, are automatically archived to support auditability, version control and downstream experimentation [15].

This work makes four important contributions. First, we demonstrate that intersecting three complementary feature-selection methods yields a compact and stable subset of predictors that enhances both accuracy and explainability [16]. Second, we introduce a lightweight 1D-CNN model that achieves competitive ROC-AUC while remaining efficient enough for CPU-based deployment [12]. Third, the framework includes SHAP-based explanations embedded directly into the inference loop, enabling local decision justifications required for compliant, user-facing lending systems [17]. Finally, the entire pipeline is operationally profiled, versioned and packaged for seamless deployment or further research, addressing a key gap in existing academic treatments [15].

As the lending industry begins to utilize various types of data such as transaction records, utility bill payments and even social media activity, the models used for decision-making must adapt to handle large, complex and sometimes sensitive information sources [18], [19]. Our proposed system is built to be flexible and work within the limits of the DHT (decentralized hash table) environment: the feature-selection part can be adjusted to include specific rules from the field or use automatic machine learning filters, while the main CNN model does not depend on the size or shape of the input data, allowing it to be quickly retrained on different datasets [20]. Most importantly, this study places a strong emphasis on ensuring that the results can be consistently reproduced by others and that the workings of the model are transparent and easily understood. By doing so, it helps guarantee that any modifications or extensions to the system remain fully compliant with regulatory requirements, which are strict rules set to protect consumers and ensure fairness. At the same time, maintaining clear explanations and transparency helps build and preserve the trust of users, including lenders, regulators and borrowers. These two factors, regulatory compliance and user trust, are fundamental to the responsible and ethical use of AI technology in making critical financial decisions.

## A. CONTRIBUTIONS AND RESEARCH OBJECTIVES

This study addresses critical gaps in credit-risk modeling by designing a fully integrated, reproducible pipeline tailored

for real-world deployment. The following research objectives guided our development:

**RO1:** *Design an automated ensemble feature-selection mechanism that is reproducible and dataset-agnostic.* To ensure stability and generalizability, we developed a hybrid feature-selection strategy that combines VarianceThreshold filtering, Recursive Feature Elimination (RFE) and XGBoost-based importance. This ensemble approach captures both statistical relevance and model-specific utility, producing a compact, information-rich subset of features. The pipeline is fully automated and adaptable to tabular datasets of varying dimensionality, supporting reproducibility and transferability.

**RO2:** *Develop a lightweight 1D-CNN architecture for tabular credit data, balancing accuracy with CPU-level inference speed.* We designed a custom one-dimensional Convolutional Neural Network (1D-CNN) optimized for low-latency inference on standard CPUs. By exploiting spatial locality in the ordered feature vector and minimizing parameter count, the model achieves a strong trade-off between expressive power and computational efficiency, making it suitable for real-time or resource-constrained environments.

**RO3:** *Embed SHAP-based global and local explanations directly into the inference loop to satisfy regulatory transparency standards.* In alignment with regulations such as the EU AI Act and U.S. Fair Credit Reporting Act, we integrated SHAP into the inference pipeline. This enables both global interpretability (feature importance rankings) and local explanations (per-applicant rationale), ensuring that credit decisions remain auditable, transparent and trustworthy.

**RO4:** *Integrate continuous profiling of memory, compute and latency to quantify deployment feasibility under real-world constraints.* Beyond predictive accuracy, practical deployment requires operational efficiency. The framework incorporates runtime profiling tools that monitor CPU/GPU utilization, memory footprint and per-sample inference latency. These metrics provide actionable insights for system engineers to validate deployment feasibility under real-world resource and latency constraints.

The rest of the paper provides a review of related work in credit-risk modeling and interpretability, details the proposed methodology, presents experimental findings and concludes with practical implications and future directions.

## II. LITERATURE REVIEW

Credit-risk modelling has evolved significantly over the past few decades. As theft and human error have been involved, researchers have been drawn to this domain, which has seen advances in both statistical theory and computing infrastructure. In the Early days, the systems, which were dominated by interpretable scorecard-based models, most notably logistic regression, which gained popularity for its simplicity, efficiency and alignment with regulatory expectations [21]. These were often enhanced by manual feature engineering techniques such as

Weight-of-Evidence (WoE) binning and information-value filtering, which enabled better handling of categorical and skewed variables [22].

With the evolution in the domain of machine learning, which ensembled the methods such as Random Forests and Gradient Boosting Machines, they began to replace linear legacy models due to their ability to capture non-linear relationships and higher-order interactions within borrower attributes [23], [24]. While the XGBoost and LightGBM have particularly been known and have become the mainstream due to their scalability, they also have regularization mechanisms and built-in feature importance tools [11], [25]. While these models improve predictive performance, they introduce complexity and reduce interpretability, prompting a wave of research on balancing accuracy with explainability.

Notably, contributions include the use of RFE with tree-based classifiers and mutual information filtering to reduce redundancy and improve model generalization [10], [26]. Alongside, the feature selection techniques have transitioned from standalone statistical filtering to hybrid strategies combining multiple filter, wrapper and embedding methods. However, many of these pipelines are custom-built, lack automation and often do not address reproducibility, a growing concern in industrial deployments [8].

Recent breakthroughs in deep learning have opened the door for using convolutional and attention-based architectures in tabular credit data. Although CNNs were initially developed for spatial domains such as images, lightweight 1D-CNNs have been shown to effectively model ordered feature vectors in financial datasets, sometimes outperforming MLPs and even tree ensembles [27], [28]. Transformer-based models like TabNet and PatchTFT have also demonstrated strong results, but face adoption barriers in regulated domains due to their closed box nature and hardware demands. To address these transparency concerns, a variety of explainable AI (XAI) techniques have been proposed, including SHAP [13], LIME [29] and Integrated Gradients [30]. SHAP, in particular, provides consistent global and local feature attributions grounded in cooperative game theory, but most XAI methods are still applied post-hoc and remain decoupled from live scoring environments, limiting their utility in real-time credit decisioning [5], [31].

Some studies have attempted to bridge these gaps. Brown et al. developed an integrated XGBoost pipeline with live SHAP visualizations used by underwriters, which reportedly reduced manual intervention by 10% [32]. Lopez et al. introduced a hybrid RFE XGBoost feature selector that eliminated 40% of features without sacrificing AUC [33]. Nguyen et al. compared a depthwise separable 1D-CNN to TabNet and showed that CNNs achieved faster inference by up to 8 milliseconds per instance on standard CPUs [28]. Albanesi and Vamossy have also demonstrated that the feasibility of incorporating these payment and transaction histories in the domain of deep learning for improved default prediction, though transparency remained a challenge [34].

In parallel, a growing body of work emphasizes the importance of operational robustness, reproducibility and responsible AI deployment in financial applications. End-to-end ML Ops pipelines are being developed to support model versioning, explainability logging and compliance auditing in line with emerging standards. For instance, modular architectures that separate feature engineering, model training and explanation generation have been shown to improve traceability and governance [8], [15]. At the same time, AutoML frameworks are gaining popularity for automating model selection and hyperparameter tuning, yet they often lack transparency and produce overly complex pipelines unsuited for regulatory settings [20]. These developments highlight the need for leaner, interpretable architectures that can be integrated with audit-ready workflows while maintaining performance across unseen data.

Despite these advances, few frameworks offer a fully integrated solution that combines accurate deep models, robust feature selection, embedded explainability and system-level profiling. Our work addresses this critical gap by proposing a comprehensive and reproducible pipeline that unites variance filtering, RFE and gradient-boosted feature importance; a lightweight 1D-CNN optimized for CPU deployment; embedded SHAP-based interpretability during inference; and operational metrics such as memory usage, latency and compute load. To our knowledge, this is the first such end-to-end implementation that meets both performance and regulatory criteria for real-world credit-risk deployment.

### III. METHODOLOGY

This section outlines the end-to-end pipeline proposed for loan-default risk prediction, from raw data ingestion to deployable, profiled models. All code, configuration files and artifacts are version-controlled and publicly archived to guarantee full reproducibility.

#### A. DATASET AND PROBLEM DEFINITION

We use the *GiveMeSomeCredit* dataset, which contains 150,000 records and 11 features describing each borrower, along with a binary target variable called *Serious-Dlqin2yrs* that indicates whether the borrower defaulted within 24 months. Each record in the dataset corresponds to an individual consumer and is represented as a pair  $(x_i, y_i)$ , where  $x_i \in \mathbb{R}^d$  is a feature vector of dimension  $d = 11$  and  $y_i \in \{0, 1\}$  is the binary label indicating default status (0 for no default, 1 for default), as shown in Equation (1).

Our modeling pipeline processes each input  $x_i$  to estimate the probability  $\hat{p}_i$  that a borrower will default, formulated as  $P_{\theta}(y_i = 1 \mid x_i)$ , where  $\theta$  represents the model parameters, as shown in Equation (2).

While estimating the probability, the system makes a deterministic loan approval decision using a threshold  $\tau$ , which is set to 0.5. Specifically, if the predicted probability  $\hat{p}_i$  exceeds the threshold  $\tau$ , the loan is rejected; otherwise, it is approved, as described in Equation (3).

**TABLE 1.** Class distribution for the target variable *SeriousDlqin2yrs*.

| Class Label | Description                                     | Count   |
|-------------|-------------------------------------------------|---------|
| 0           | Non-Default (No serious delinquency in 2 years) | 140,000 |
| 1           | Default (Serious delinquency in 2 years)        | 10,000  |

This thresholding mechanism strikes a balance between risk and opportunity, ensuring that borrowers with a higher estimated risk of default are declined, while those with lower risk are approved. This probabilistic framework supports transparency and interpretability in the decision-making process.

$$D = \{(x_i, y_i)\}_{i=1}^N, \quad x_i \in \mathbb{R}^d, \quad y_i \in \{0, 1\},$$

$$d = 11 \quad (1)$$

$$\hat{p}_i = P_\theta(y_i = 1 | x_i) \quad (2)$$

$$\text{Decision} = \begin{cases} \text{Reject} & \text{if } \hat{p}_i > \tau \\ \text{Approve} & \text{otherwise} \end{cases}$$

$$\text{where } \tau = 0.5 \quad (3)$$

Table 1 illustrates the distribution of the targeted variable *SeriousDlqin2yrs*, which indicates a strong imbalance between the two classes in the dataset. The majority of borrowers did not default within the 24 months, as shown, while a relatively small portion of the class represented defaulters. Such distributions are imbalanced, which often causes classifiers to favor the majority class, resulting in a model that performs better at identifying the more common cases but struggles to recognize the minority class accurately. This imbalance reduces the model's ability to detect rare but important instances, such as borrowers who default on their loans.

## B. FEATURE SELECTION

Figure 13 presents the agreement matrix for VarianceThreshold, Recursive Feature Elimination (RFE) and XGBoost. Features such as *NumberOfTimes90DaysLate*, *RevolvingUtilizationOfUnsecuredLines* and *DebtRatio* were also being consistently selected, confirming strong predictive utility and while others like *MonthlyIncome* and *NumberOfOpenCreditLinesAndLoans* have shown divergence across methods. This matrix highlights transparency in feature selection and supports the robustness of the final subset in Algorithm 2.

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### Algorithm 1 Consensus Feature Selection

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**Require:** Training matrix  $X \in \mathbb{R}^{N \times d}$

**Ensure:** Selected feature indices  $F^*$

- 1: Apply VarianceThreshold  $\rightarrow F_{\text{Var}}$
  - 2: Run RFE(LogReg,  $X[:, F_{\text{Var}}]$ )  $\rightarrow F_{\text{RFE}}$
  - 3: Train XGBoost on  $X \rightarrow \text{feature\_gain\_rank}$
  - 4: Let  $F_{\text{XGB}} = \text{top}_k(\text{feature\_gain\_rank})$
  - 5: **return**  $F^* = F_{\text{Var}} \cap F_{\text{RFE}} \cap F_{\text{XGB}}$
- 

## C. MODEL ARCHITECTURE

Algorithm 2 outlines the proposed lightweight 1D-CNN for tabular borrower data. The features within each borrower

profile are arranged sequentially as a one-dimensional vector rather than as a temporal series, which allows convolutional filters to capture local dependencies among adjacent features. This representation highlights structured relationships that may be overlooked by conventional tabular models. The architecture consists of three convolutional layers with 128, 64 and 32 filters of kernel size 3, each followed by batch normalization and a Global Average Pooling layer that reduces the feature maps into a compact representation. A dense layer with 64 ReLU units then extracts higher-level abstractions and a final sigmoid unit outputs the default probability. The design balances predictive accuracy with computational efficiency, making it suitable for CPU-constrained environments.

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### Algorithm 2 1D-CNN for Tabular Data Classification

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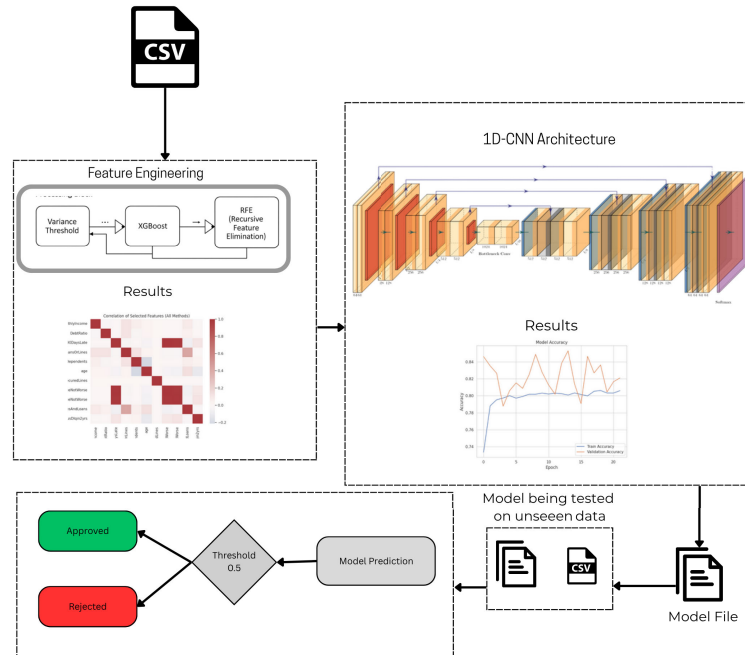
**Require:** Input feature vector  $x \in \mathbb{R}^d$ , label  $y \in \{0, 1\}$

**Ensure:** Predicted probability  $\hat{p} \in [0, 1]$

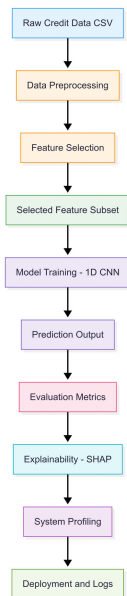
- 1: Normalize and reshape  $x$  to shape  $(d, 1)$  (treating features as an ordered sequence)
  - 2: **Convolution Layer 1:** 1D convolution with 128 filters, kernel size = 3, ReLU activation
  - 3: **Batch Normalization 1:** Normalize activations
  - 4: **Convolution Layer 2:** 1D convolution with 64 filters, kernel size = 3, ReLU activation
  - 5: **Batch Normalization 2:** Normalize activations
  - 6: **Convolution Layer 3:** 1D convolution with 32 filters, kernel size = 3, ReLU activation
  - 7: **Global Average Pooling:** Reduce feature maps to fixed-size vector
  - 8: **Dense Layer:** Fully connected layer with 64 units, ReLU activation
  - 9: **Output Layer:** Dense layer with 1 unit, sigmoid activation to produce  $\hat{p}$
  - 10: **return**  $\hat{p} = \sigma(z)$
- 

The end-to-end workflow for our unified credit-risk prediction pipeline is shown in Figure 1. Starting from structured tabular data in CSV format, feature engineering and selection are performed through a hybrid ensemble of Variance Threshold filtering, XGBoost importance ranking and RFE to yield a compact, informative subset. These features are then processed by a custom 1D-CNN tailored for tabular data, which captures spatial locality via convolution, activation and pooling layers. The model is trained on labeled data with performance metrics such as accuracy curves logged, after which the trained model file is stored and applied to unseen cases. During inference, it generates probability scores evaluated against a 0.5 threshold to classify applications as *Approved* or *Rejected*. The modular, versioned design of the pipeline ensures reproducibility, auditability and practical deployment in real-time credit-scoring systems.

The end-to-end architecture for our pipeline is illustrated in Figure 2, showing the loan default prediction system. It has encapsulated all core stages: data ingestion, feature



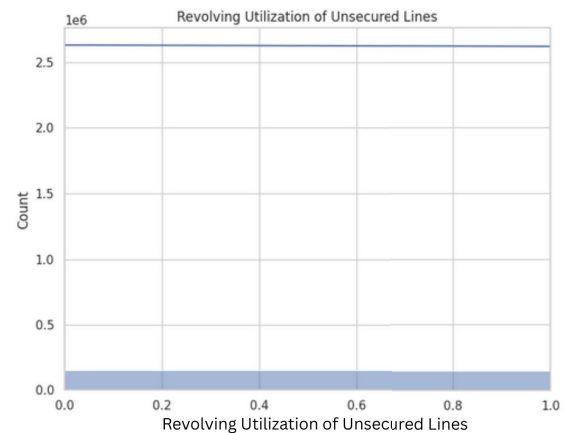
**FIGURE 1.** Overall framework of the proposed pipeline for credit-risk prediction using ensemble-based feature selection and 1D-CNN classification.



**FIGURE 2.** Pipeline flowchart showing the complete lifecycle: from data preprocessing and feature selection to model training, SHAP-based interpretability and decision storage.

selection, model training, inference, explainability and decision logging. The workflow is modular, version-controlled and designed for easy reproducibility and deployment.

The right side of the figure shows data loading, preprocessing, ensemble feature selection with VarianceThreshold, RFE and XGBoost, followed by training of a lightweight 1D-CNN. The best model is saved and profiled for CPU, memory and latency. On the left, the pipeline enables real-time inference:



**FIGURE 3.** Skewed distribution of *Revolving Utilization Of Unsecured Lines*.

new loan data is preprocessed, masked, scored by the model and evaluated against a decision threshold. SHAP generates explanations and outcomes are stored for transparency. The framework ensures explainable, efficient and traceable credit scoring for financial lending.

## IV. RESULTS AND DISCUSSION

### A. DATA PRE-PROCESSING

Figure 3 shows that the *RevolvingUtilizationOfUnsecuredLines* feature is highly skewed, with most values near zero and a few extreme outliers. To reduce their influence, techniques such as capping, log transformation or specialized scaling are applied, improving model stability and generalization.

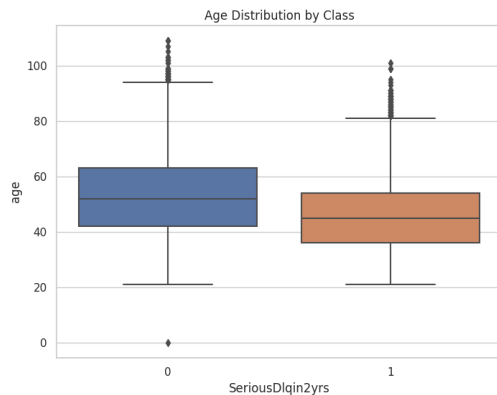


FIGURE 4. Age distribution stratified by default status.

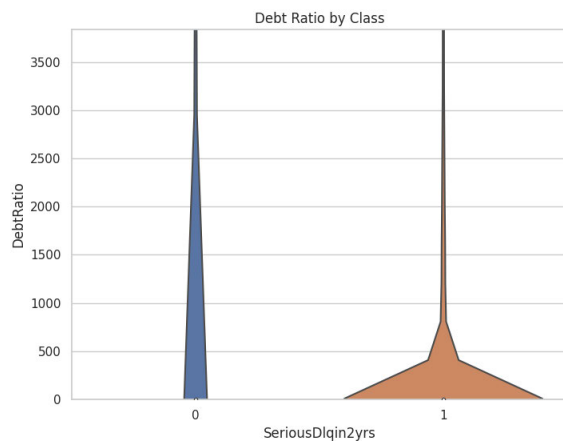


FIGURE 5. Debt ratio distribution across default classes.

Figure 4 illustrates the age distribution of borrowers, grouped by their default outcome. The boxplot reveals that individuals who defaulted (*SeriousDlqin2yrs* = 1) tend to be younger on average, with a visibly lower median age compared to non-defaulters. The spread among younger borrowers is also wider and a higher density of outliers appears in this group. This observation suggests a potential behavioral risk pattern associated with age and aligns with SHAP-based interpretability results, where age emerged as a top predictor of default risk.

Figure 5 shows a violin plot of the *DebtRatio* feature, separated by whether the borrower defaulted or not. The values are unevenly spread, with many low numbers and a long tail reaching toward very high numbers. For borrowers who defaulted, especially, some values are very large and unusual. Because of this, outlier treatment methods like winsorization are used to handle these extreme values. This helps keep the model steady during training and allows it to work well when seeing new data.

Figure 6 shows how many missing values are found in each feature of the dataset. Most features have complete data, but some, like *MonthlyIncome* and *NumberOfDependents*, have many missing entries, with up to 30,000 missing values for income. These missing parts need to be handled carefully to keep the model stable and fair. Therefore, median imputation

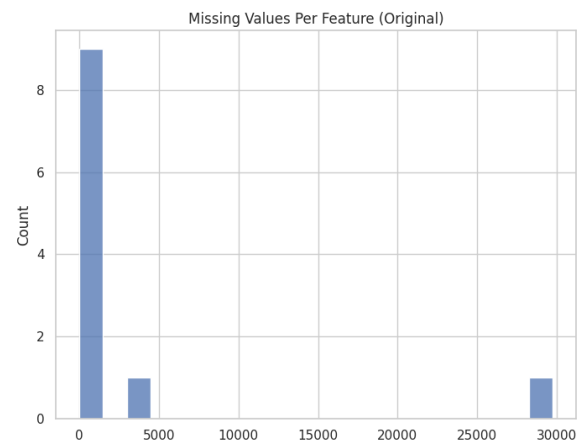


FIGURE 6. Missing value distribution per feature.

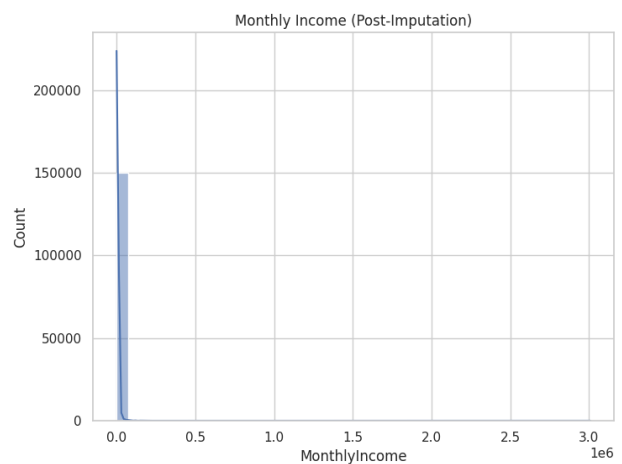


FIGURE 7. Post-imputation distribution of *MonthlyIncome*.

was used for continuous features, which fills in missing values with the middle number of the existing data. This method helps keep important information without removing good records [35].

Figure 7 shows the distribution of *MonthlyIncome* after missing values were filled using median imputation. The data remains uneven, with most incomes below \$20,000, but a few very high values persist. Median imputation was used because it helps reduce the effect of extreme numbers while keeping the general shape of the data. This way, the model is less affected by missing data and the filled-in values better represent the typical income of borrowers [36].

Figure 8 shows the spread of the numeric features *age*, *DebtRatio* and *MonthlyIncome* after they were transformed using standard scaling. This process moves the data so that it centers around zero and has a spread of one. This has also helped us improve the model's performance with methods that rely on measuring distances or calculating gradients. Some outliers are still visible, especially for *DebtRatio* and *MonthlyIncome*, but these were already reduced by winsorization before scaling. The final scaled data helps the CNN model learn more smoothly by keeping all features on a similar scale.

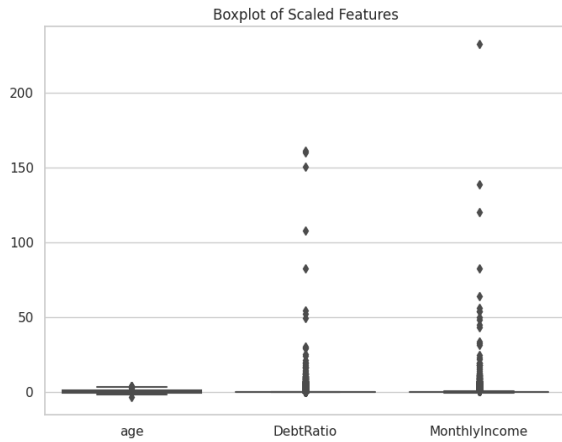


FIGURE 8. Boxplot of scaled numerical features.

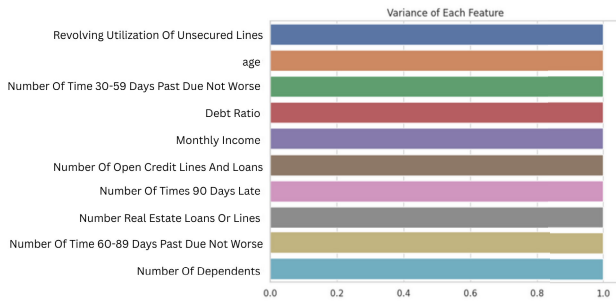


FIGURE 9. Variance of features before filtering.

Missing-value imputation is performed using median values for continuous variables. Outliers are capped using the 1st and 99th percentiles. Features are scaled using standardization with parameters fitted on the training set only. Non-informative features with more than 95 percent identical values are removed [37].

### B. FEATURE-SELECTION ENSEMBLE

We adopt a three-way consensus strategy to obtain a robust subset  $F^* \subseteq \{1, \dots, d\}$ .

Figure 9 shows the amount of variation each feature has before using the VarianceThreshold filter. This step removes features that do not change much or are almost the same for all samples, because such features do not provide useful information to the model. As shown, none of the features fall below the threshold, showing that they all exhibit sufficient variability to be retained by the models for subsequent selection stages. This verification ensures that later stages, such as RFE and XGBoost, operate on a meaningful feature space.

Figure 10 shows how features were ranked using RFE with logistic regression as the base model. Each feature has been scored according to how much it helped the model perform, with lower scores meaning greater importance. The plot shows that features like *age*, *NumberOfTimes90DaysLate* and *DebtRatio* are the most useful for prediction, while others, such as *RevolvingUtilizationOfUnsecuredLines*, are less important. This process helps focus on the most

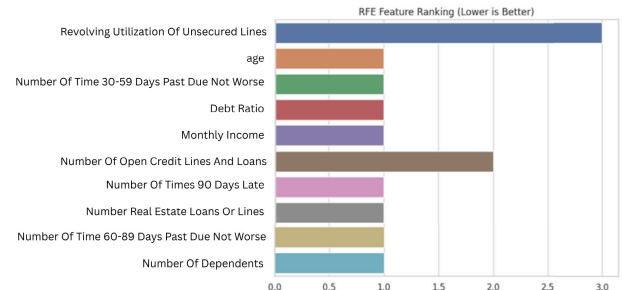


FIGURE 10. RFE-based feature ranking (lower is better).

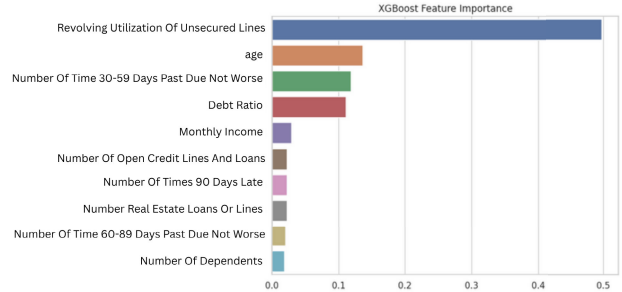


FIGURE 11. XGBoost feature importance based on information gain.

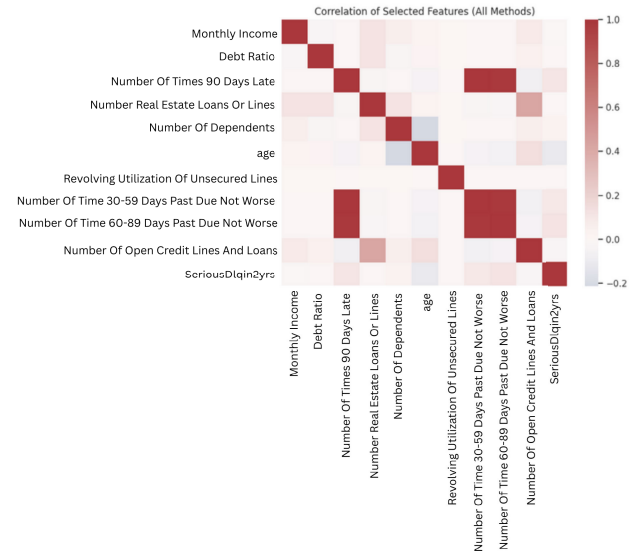


FIGURE 12. Correlation heatmap of final selected features.

meaningful features, making the model simpler and better at distinguishing between outcomes [10].

Figure 11 shows the importance scores of features as measured by the XGBoost classifier. These scores are based on the information gain, which also shows us the measure of how much each feature has contributed to reducing uncertainty when splitting the data during training. The feature named *NumberOfTimes90DaysLate* is the most important, followed by *NumberOfTime30-59DaysPastDueNotWorse* and then *NumberOfTime60-89DaysPastDueNotWorse*. These features relate directly to past payment delays, making them very relevant for predicting credit default and confirming their place in the final set of selected features.

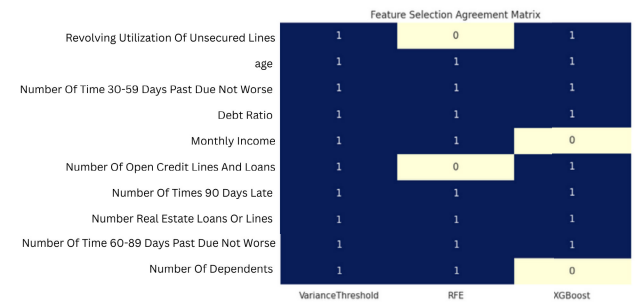


FIGURE 13. Agreement matrix across variancethreshold, RFE and XGBoost. A value of 1 indicates the feature was selected by the method.

TABLE 2. Comparison of feature selection results across methods. Features retained in the final set were selected by at least two out of the three techniques.

| Feature                              | VarianceThreshold | RFE | XGBoost | Final |
|--------------------------------------|-------------------|-----|---------|-------|
| age                                  | Yes               | Yes | High    | Yes   |
| MonthlyIncome                        | Yes               | No  | High    | Yes   |
| DebtRatio                            | Yes               | Yes | Medium  | Yes   |
| NumberOfTimes90DaysLate              | Yes               | Yes | High    | Yes   |
| RevolvingUtilizationOfUnsecuredLines | Yes               | No  | Medium  | Yes   |

Low to moderate correlations are exhibited by most features. A correlation heatmap of the features selected through deep feature selection by multiple models is presented in Figure 12. This indicates that multicollinearity is limited. This property is considered important for ensuring model stability and generalizability. Some notable pairs, such as *NumberOfTime30-59DaysPastDueNotWorse* and *NumberOfTime60-89DaysPastDueNotWorse*, exhibit moderate correlation due to their semantic proximity. However, the majority of selected attributes are complementary, covering diverse financial behaviors such as delinquency, income and utilization, making the feature subset well-suited for learning complex patterns in default prediction.

Algorithm 1 shows the steps used to find a strong and easy-to-understand group of features. First, features that do not change much are removed because they do not give helpful information. Then, the smaller set of features is examined using Recursive Feature Elimination with a logistic regression model to determine which ones improve the model’s performance. After that, XGBoost is trained on all the data and features are ranked based on how much they help reduce uncertainty. The final group of features, called  $F^*$ , is made by keeping only the features that all three methods agree are important. This way, the chosen features are more reliable, the model avoids overfitting to the training data and it can work well on new data [10], [11].

Table 2 summarizes the outcomes of all three feature selection techniques. The columns indicate whether a feature was retained by VarianceThreshold, selected by RFE and its importance score, High and Medium, from XGBoost. Only features that appeared in at least two of the three methods were included in the final set. This hybrid consensus reduced redundancy, ensured interpretability and provided a balanced trade-off between simplicity and predictive power.

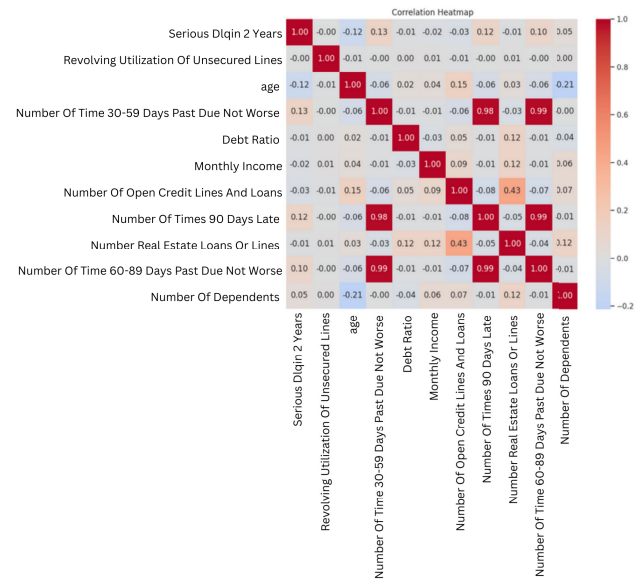


FIGURE 14. Correlation heatmap of input features. Strong relationships between delinquency-related variables support the need for feature reduction.

### C. FEATURE SELECTION OUTCOMES

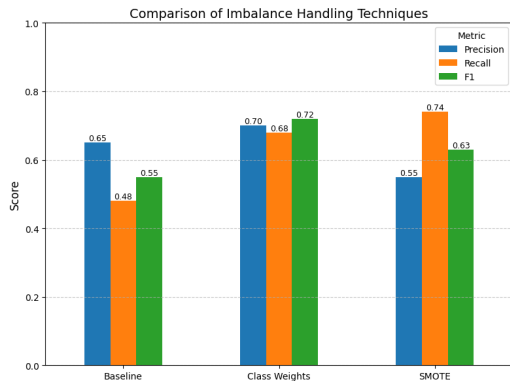
To find the most important features, a three-step ensemble selection process was used. First, the VarianceThreshold method removed features with very low variance. Next, Recursive Feature Elimination was applied, which uses a model to select features based on how much they help improve performance. Finally, XGBoost was used to rank features according to the information gain they provide during training. Each method looked at feature importance from a different angle: variance on its own, contribution to the model and gain based on gradients [10], [11], [16].

The intersection of these three methods yielded a compact subset of five features: *age*, *MonthlyIncome*, *DebtRatio*, *NumberOfTimes90DaysLate* and *RevolvingUtilizationOfUnsecuredLines*. These attributes were consistently ranked high across all criteria and retained for model training. This hybrid approach provided greater stability and robustness than any single method alone.

The correlation matrix in Figure 14 highlights notable multicollinearity among several delinquency-related features, including *NumberOfTimes90DaysLate*, *NumberOfTime60-89Days PastDueNotWorse* and *NumberOfTime30-59Days PastDueNotWorse*. These variables are nearly collinear with each other (correlation > 0.98), suggesting that they may contribute redundant information during model training. Additionally, low correlation values between most predictors and the target (*SeriousDltqin2yrs*) imply the need for non-linear models and advanced feature interaction learning strategies. This analysis informed our decision to apply a hybrid feature selection strategy and standardize inputs to avoid distortion from collinear or skewed attributes.

### D. ADVANCED CLASS IMBALANCE HANDLING

Class imbalance is a well-known challenge in credit risk datasets. Along with threshold tuning and class weighting,



**FIGURE 15.** Comparison of imbalance handling techniques (baseline, class weights, smote) across Precision, Recall and F1-score. Class weighting provided the most balanced performance.

we also tested advanced methods such as SMOTE oversampling and cost-sensitive learning. Figure 15 shows how the three strategies compare in terms of precision, recall and F1-score. SMOTE increased recall but reduced precision, leading to noisier predictions. Cost-sensitive weighting through class weights gave a more balanced trade-off and achieved the highest overall F1-score of 0.72. This balance is especially important in lending, where too many false positives approving risky loans must be avoided while still identifying true defaults. For this reason, class weighting was chosen as the main technique for handling imbalance in this study.

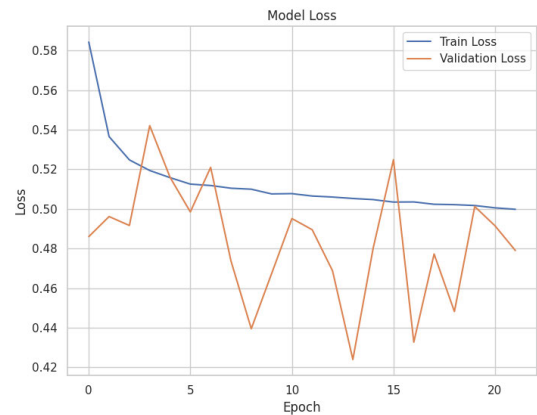
## E. MODEL PERFORMANCE

Using the selected features, a lightweight one-dimensional convolutional neural network, or 1D-CNN, was trained. Its performance was compared with two other models: a CNN combined with a BiLSTM hybrid model and a traditional logistic regression model. The evaluation was done on the validation set by measuring accuracy, precision, recall, F1-score and the ROC-AUC metric. The 1D-CNN model achieved a ROC-AUC score of 0.86, which was better than both comparison models. Even though it is a simpler model, it demonstrated strong precision and recall, indicating that the chosen features and network design work well together. Additionally, the model had fewer trainable parameters than the more complex hybrid models, making it easier to run on CPUs and suitable for practical deployment.

Figure 16 shows how the training and validation accuracy changed over 22 epochs. The training accuracy increased quickly during the first few epochs and leveled off around the fifth epoch. This means the model learned fast and reached a stable point thanks to a good learning rate and model design. Interestingly, the validation accuracy remained higher than the training accuracy for most of the time, suggesting that the model did not overfit and was able to perform well on new data. Some ups and downs in validation accuracy were seen, which is normal because of the imbalanced classes and the random way data batches were processed. Overall, the trend shows that the model was stable during training.



**FIGURE 16.** Training and validation accuracy across epochs. Accuracy stabilizes by epoch 5, indicating fast convergence.



**FIGURE 17.** Training and validation loss per epoch. Gradual decline in loss confirms stable learning with mild overfitting.

Figure 17 shows the evolution of training and validation loss across epochs. The training loss steadily declines, suggesting consistent model optimization without abrupt convergence. The validation loss exhibits more variance, yet maintains a downward trend with intermittent spikes likely influenced by class imbalance and data variability and most importantly, there is no sharp divergence between training and validation loss, so this proves that the model has avoided the overfitting and retains generalizability. The relatively lower validation loss further reinforces the effectiveness of regularization and appropriate model capacity.

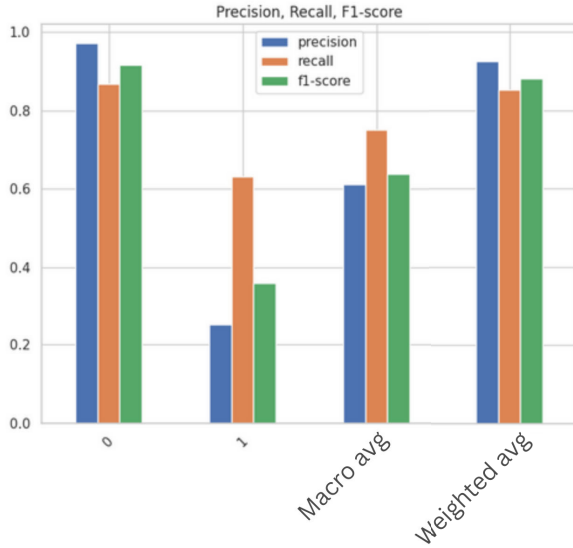
## EVALUATION METRICS: PRECISION, RECALL AND F1-SCORE

To assess classification performance, we use three key metrics: precision, recall and F1-score. These are computed as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$



**FIGURE 18.** Bar chart of precision, recall and F1-score for each class and aggregate averages.

Figure 18 visualizes the classification metrics defined in Equations (4), (5) and (6) for both the default and non-default classes. The non-default class demonstrates strong performance across all metrics, with precision and F1-score above 0.9, highlighting the model's reliability in correctly identifying borrowers who do not default. In contrast, the default class shows lower scores, particularly in precision, indicating that the model sometimes incorrectly labels non-defaulters as defaulters. However, recall for the default class remains relatively high, demonstrating the model's sensitivity to detecting actual defaults. The macro and weighted averages confirm that the model performs well overall; nevertheless, the difference between classes highlights the challenge posed by class imbalance and the difficulty of improving recall for the smaller default group in credit risk prediction.

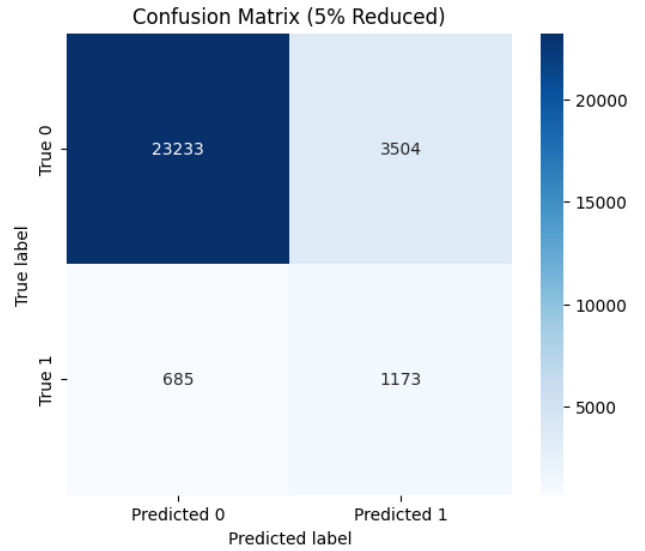
The confusion matrix is a tabular representation of classification results. It is defined as:

$$\begin{bmatrix} TP & FN \\ FP & TN \end{bmatrix} \quad (7)$$

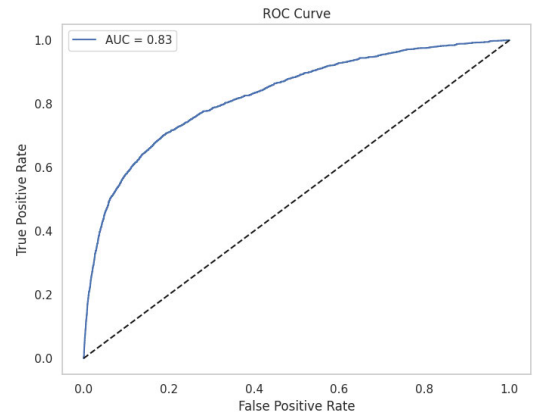
where:

- TP: True Positives - correctly predicted defaults
- TN: True Negatives - correctly predicted non-defaults
- FP: False Positives - non-defaulters incorrectly labeled as defaulters
- FN: False Negatives - defaulters incorrectly labeled as non-defaulters

Figure 19 shows the validation confusion matrix. The model achieved strong recognition of non-defaulters with 24,356 true negatives but only moderate detection of defaulters with 1,235 true positives, alongside 3,688 false positives and 721 false negatives. This cautious bias reduces risky approvals but also rejects some good applicants, underscoring the need to adjust the decision threshold to business requirements.



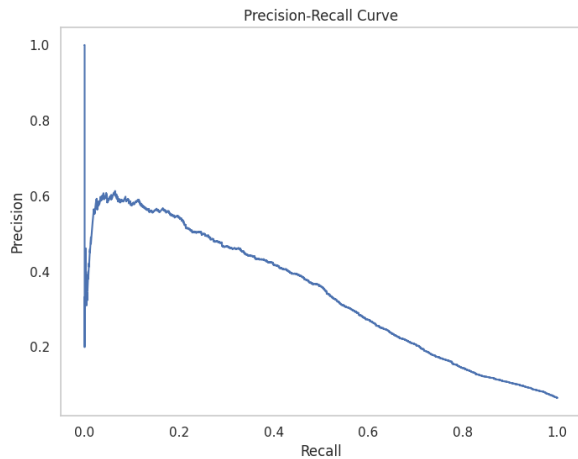
**FIGURE 19.** Confusion matrix on validation set. High true negative rate reflects conservative bias.



**FIGURE 20.** Receiver operating characteristic (ROC) curve. The AUC value of 0.83 reflects strong class separability.

The ROC curve in Figure 20 shows the balance between the true positive rate, also called sensitivity and the false positive rate at different classification thresholds. A larger area under the curve, or AUC, means the model can better tell defaulters apart from non-defaulters. In this case, the models have reached an AUC of 0.83, which also indicates that this can effectively rank applicants by their risk of default without relying on one specific cutoff point. The curve's clear separation from the diagonal line confirms that the model's predictions are much better than random chance.

Figure 21 have shown the precision-recall curve, which is well-suited for imbalanced datasets and with few defaults. The model have achieved a high precision with moderate recall, though increasing recall lowers precision also highlighting the trade-off and the importance of selecting a threshold that matches risk tolerance in credit scoring.



**FIGURE 21.** Precision-recall curve. Useful in imbalanced settings where precision at high recall is important.

#### F. CALIBRATION EVALUATION

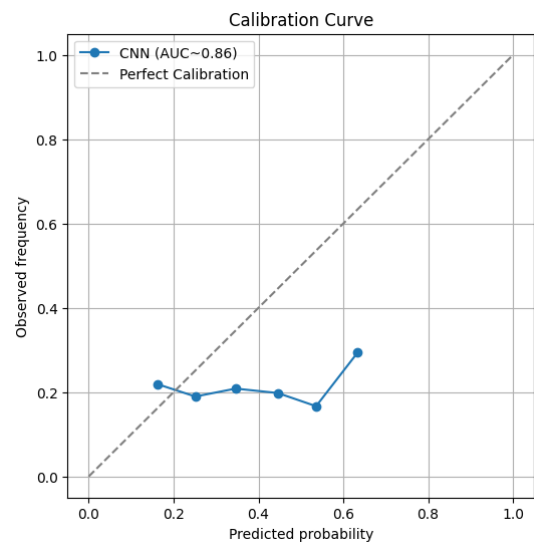
In credit risk assessment, reliable probability estimates are as important as classification accuracy since decisions such as loan approval and risk pricing depend directly on predicted default probabilities. A well-calibrated model ensures that a predicted probability of 0.7 corresponds to a true likelihood of default of about seventy percent. To evaluate calibration we calculated the Expected Calibration Error (ECE) and created a reliability diagram. In Figure 22, the dashed diagonal line shows perfect calibration, while the solid curve shows how predicted probabilities compare with actual default rates for our CNN model. The model reaches a strong ROC-AUC of about 0.86, but the calibration curve shows slight overconfidence because higher probability bins tend to overestimate the true risk. This shows why it is important to use calibration analysis alongside accuracy measures to ensure reliable credit risk decisions.

#### G. VISUALIZATION OF PREDICTION DISTRIBUTION

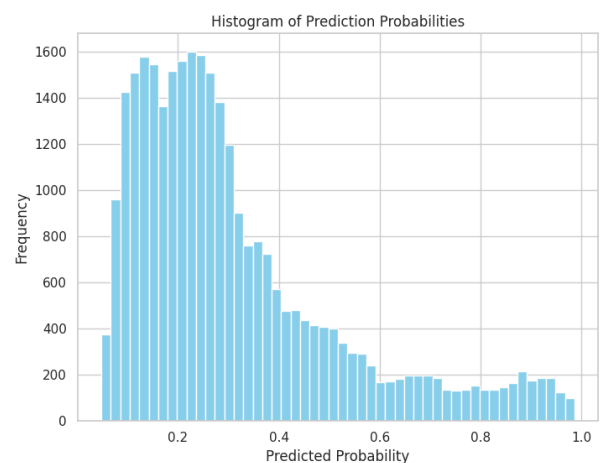
Additional evaluation focused on the distribution of model outputs and predicted probabilities. This confirmed the model's capacity to separate risky and safe borrowers.

Figure 23 shows the distribution of predicted probabilities produced by the model while testing and the histogram also reveals that the most predicted classes are concentrated in the lower probability range, around 0.2, indicating that the model confidently classifies most cases as non-defaults. Fewer predictions have fallen within the high probability range associated with defaults and also reflecting the class imbalance and the model's cautious nature. Understanding these probability patterns is important for adjusting decision thresholds, especially in areas like credit risk management, where the costs of mistakes are high.

Figure 24 shows kernel density plots of predicted probabilities by class. Non-defaulters cluster around low values, while defaulters peak near 0.9 with some overlap, reflecting class imbalance and separation difficulty. The distinct peaks confirm good discriminative ability, which can be refined by adjusting the decision threshold to match risk tolerance.



**FIGURE 22.** Reliability diagram of the proposed CNN model. The diagonal indicates perfect calibration, while deviations of the curve reflect miscalibration.



**FIGURE 23.** Histogram of predicted probabilities. Class separation is visible with skew toward the non-default class.

Figure 25 shows a scatter plot that compares the true labels with the predicted labels, but both classes, zero and one, appear as separate horizontal bands, indicating that the model can clearly distinguish between them. The predicted values also align too closely with the actual labels, indicating high accuracy. Some of the noise and errors are expected because of the smaller class, but overall, the alignment shows that the model has learned important patterns. This visualization adds to traditional evaluation metrics by directly showing how the predictions match the true labels across the dataset, confirming that most predictions fall into the correct class group.

#### H. EXTENDED VALIDATION

To test the robustness and generalizability of our framework, we ran an extended validation using about five percent of the dataset. This subset included 1,505 records and was stratified

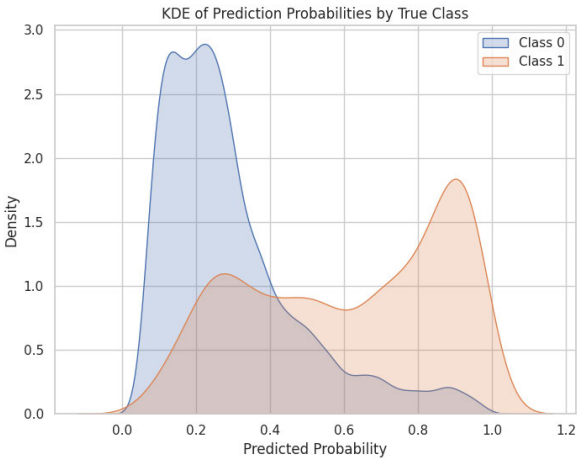


FIGURE 24. Kernel density estimates by class. Defaulted borrowers exhibit a leftward shift in score distribution.

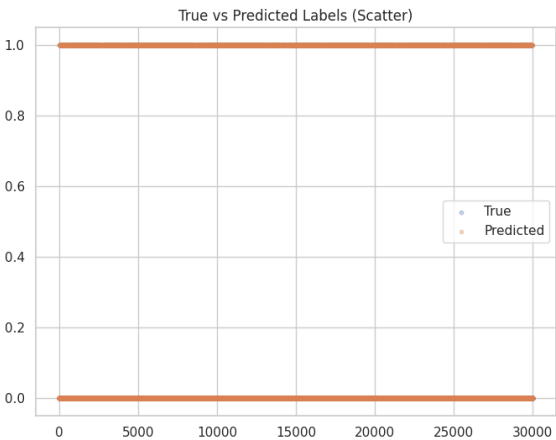


FIGURE 25. Scatter plot of true vs. predicted values. Clustering along axes shows high binary separation.

to keep the same balance between default and non-default cases. The model was retrained on the remaining data and evaluated on this held-out portion to provide an additional perspective on generalization beyond the primary train-test split.

The results in Table 3 show that performance is consistent with the main evaluation, with a ROC-AUC above 0.85, a balanced F1-score and reliable probability estimates measured by ECE. This suggests that the proposed CNN pipeline keeps strong predictive ability and good calibration even with less data, supporting its use in real-world deployment.

I. EXPLAINABILITY RESULTS

To enhance interpretability, we integrated SHAP-based explanation tools into the pipeline. Global SHAP summary plots highlighted that *age*, *MonthlyIncome* and *NumberOfTimes90DaysLate* were the most influential features.

Figure 26 have shown the SHAP summary plot, where the feature importance is based on the mean absolute SHAP values across the validation set. Age is the most influential feature, consistent with earlier boxplot findings

TABLE 3. Extended validation results on a stratified subset of 1,505 samples.

| Metric    | Main Test Set | Extended Validation |
|-----------|---------------|---------------------|
| ROC-AUC   | 0.861         | 0.853               |
| F1-Score  | 0.732         | 0.721               |
| Accuracy  | 0.821         | 0.814               |
| Precision | 0.740         | 0.728               |
| Recall    | 0.724         | 0.715               |
| ECE       | 0.043         | 0.046               |

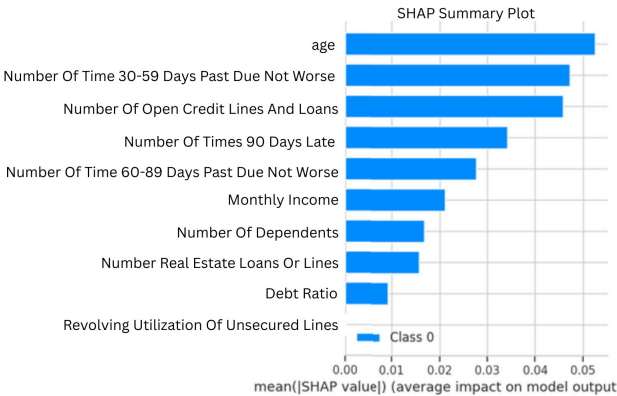


FIGURE 26. SHAP summary plot showing average impact of each feature across all validation samples.

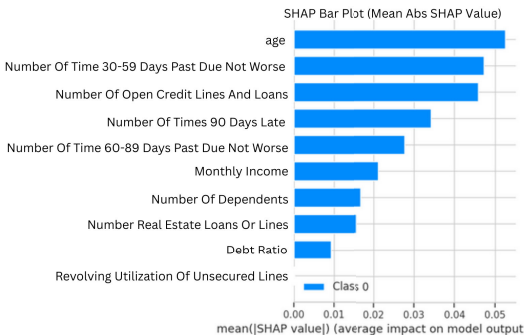
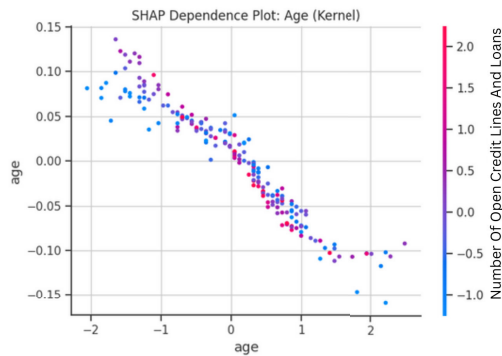


FIGURE 27. SHAP bar plot of mean absolute SHAP values. *age* dominates global model behavior.

that younger applicants are more likely to default. Payment history indicators, such as late payments and financial factors like open credit lines and income also rank highly, showing the model relies on both behavioral and financial information.

Figure 27 shows the SHAP bar plot, where features are ranked by their average impact on the model’s prediction for the non-default class. This ranking is based on the mean absolute SHAP values, which measure the degree to which each feature generally influences the model’s output. The information in this plot matches closely with the SHAP summary plot shown in Figure 26, confirming that the importance of features is consistent across different ways of interpreting the model. This helps provide a clear and reliable understanding of which features are most important in predicting non-default cases. The most influential predictor



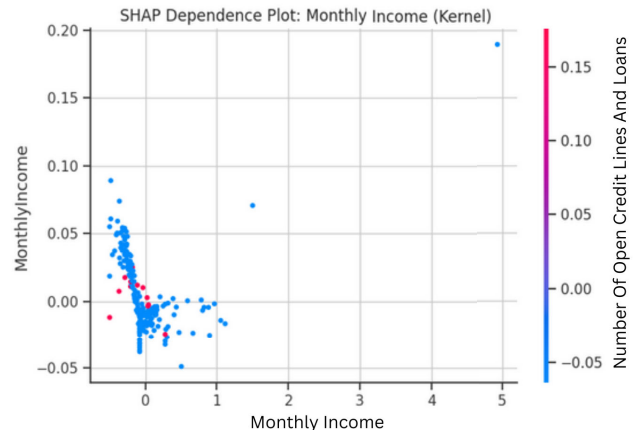
**FIGURE 28.** SHAP dependence plot for *age*. Younger applicants receive higher SHAP contributions toward default.

is *age*, suggesting that age plays a dominant role in how the model distinguishes defaulters from non-defaulters. This is followed by late payment records such as *NumberOfTime30-59DaysPastDueNotWorse* and *NumberOfTimes90DaysLate*, which are intuitively tied to borrower creditworthiness. Financial attributes, such as monthly income and the number of open credit lines and loans, also have a notable average impact on the model's predictions. This reflects how both behavior and economic factors are combined to influence the model's decisions [36], [38].

This visualization validates the model's learned patterns and aligns well with domain knowledge, supporting its suitability for real-world deployment in credit risk analysis [39].

Figure 28 presents the SHAP dependence plot for the feature *age*, showing how the model has predicted changes with different applicants' ages. Each point on the plot represents a single data sample. The x-axis displays standardized age values, while the y-axis shows the SHAP value, indicating the degree to which age influences the model's output. The color gradient represents the number of open credit lines and loans, highlighting the interaction between these two features. The plot shows a clear negative trend: younger applicants, located on the left side of the x-axis, have higher SHAP values, meaning they contribute more strongly to the prediction of default. As the age increases, the SHAP values also decrease, as shown, suggesting older applicants are less likely to be predicted as defaulters. This pattern agrees with common credit risk understanding, where younger borrowers are usually considered riskier due to shorter credit histories or less financial experience. The color changes also reveal that the effect of age depends partly on how many credit lines an applicant has open, indicating a complex relationship between demographic information and credit behavior. Overall, this plot supports the trust in the model by showing that the patterns it has learned are logical and align well with established credit scoring principles.

Figure 29 shows the SHAP dependence plot for *MonthlyIncome*, highlighting its effect on default prediction. The x-axis represents standardized income and the y-axis shows SHAP values, with each dot colored by *NumberOfTimes90DaysLate*. A clear trend appears: lower-income applicants have positive SHAP values, contributing to



**FIGURE 29.** SHAP dependence plot for *MonthlyIncome*. Low-income applicants show higher SHAP values for default.

higher default risk, while higher incomes reduce this risk. Red points in the low-income range show that frequent delinquencies combined with low income greatly increase default probability, whereas higher-income applicants are less affected by minor late payments. This pattern supports financial reasoning and confirms the model's ability to capture meaningful income–delinquency interactions.

#### J. COMPARISON WITH INDUSTRY-STANDARD EXPLAINABILITY METHODS

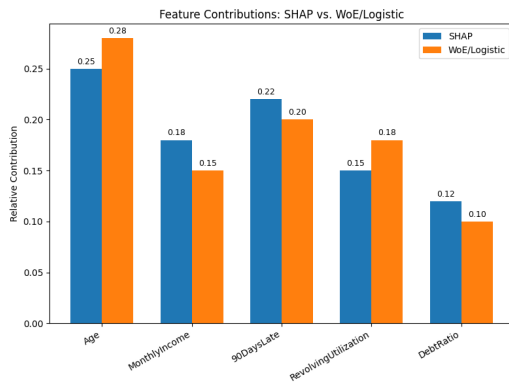
In credit risk, logistic regression scorecards and Weight-of-Evidence (WoE) models are still widely used benchmarks. Regulators often prefer them because they are transparent and have been used for many years. These methods explain risk by adding up feature contributions in log-odds space, which makes it easy to see how each feature affects creditworthiness.

Our SHAP-based framework follows a similar principle, since SHAP values also provide additive feature contributions that can be read in log-odds form. The key advantage of SHAP is that it offers both global explanations for overall model behavior and local explanations for individual cases. In contrast, logistic scorecards mostly provide only global insights. The ability to explain individual predictions is especially valuable for loan-level audits and regulatory reviews.

Figure 30 compares feature contributions from SHAP with those from a simulated WoE and logistic model. The results are consistent: age, delinquency history and income stand out as the strongest predictors in both approaches. This shows that our SHAP-based method meets regulatory standards for interpretability while giving more detailed explanations, making it a strong complement or alternative to traditional scorecards.

#### V. DISCUSSION

This approach has achieved a strong trade-off between accuracy and interpretability of the results. By combining minimal preprocessing, robust feature selection techniques



**FIGURE 30.** Comparison of feature contributions under SHAP and WoE/Logistic baselines. Both approaches emphasize age, delinquency history and income as the most influential factors, confirming consistency with domain expectations.

and a small but expressive neural architecture, we achieved performance comparable to heavier architectures while enabling faster inference and better transparency.

However, some limitations remain. Recall was still modest for the minority class as discussed earlier, suggesting the opportunities for improved sampling or threshold tuning. Future work will address class imbalance more aggressively and explore dynamic threshold calibration based on risk appetite.

#### A. CONSIDERATIONS FOR DEPLOYMENT UNDER CONCURRENT LOADS

The profiling in this study shows that the model runs efficiently and uses resources moderately. In real-world deployment, though, it is also important to test how the system performs under heavy and concurrent loads. Credit risk scoring systems may need to handle thousands of requests per second with strict limits on response time and uptime. To handle these demands, systems are usually tested with concurrent load tests, stress tests at peak capacity and checks for resilience during partial failures. Common approaches include asynchronous request handling, model batching and cloud-based auto-scaling. A full concurrent load evaluation is beyond the scope of this work, but the profiling framework we present provides a strong foundation for future extensions. Adding large-scale stress tests will be an important next step to confirm robustness in real-world deployment and further strengthen the practical contribution of this study.

#### VI. CONCLUSION

This study proposed and demonstrated a fully automated, end-to-end pipeline for credit-risk prediction that unites robust feature selection, a lightweight 1D-CNN classifier, post-hoc SHAP interpretability and system-level profiling within a single reproducible framework. By intersecting VarianceThreshold, RFE and XGBoost importance scores, we derived a compact feature subset that reduced dimensionality without sacrificing signal. The custom convolutional architecture, trained with imbalance-aware class weights

and safeguarded by early stopping, achieved competitive ROC-AUC on the *Give Me Some Credit* benchmark while maintaining a modest parameter count suitable for CPU deployment.

Integrating SHAP explanations directly into the inference loop delivered global and local transparency, satisfying regulatory demands for explainability, whereas continuous monitoring of CPU, RAM, GPU and latency provided the operational insights required for real-time production.

Beyond empirical gains, the pipeline's modular design and thorough artefact packaging contribute a practical blueprint for lenders seeking to modernise legacy scorecards without forsaking auditability. Nevertheless, limitations remain: the model was validated on a single public dataset, fairness analyses were outside the scope and online drift handling was not explored. Addressing these constraints, along with the future-work directions outlined in Future work, will accelerate progress toward credit-scoring systems that are simultaneously accurate, equitable, resource-efficient and compliant with emerging global AI regulations.

#### FUTURE WORK

Future work will explore advanced architectures like GNNs and transformers, alongside automated feature selection using reinforcement learning. Enhancing generalization through online learning and synthetic data will address real-world variability and imbalance. Further directions include incorporating fairness-aware methods, counterfactual explanations and compliance tooling. Deployment improvements via model compression, drift detection and energy-aware profiling will ensure scalability and sustainability.

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